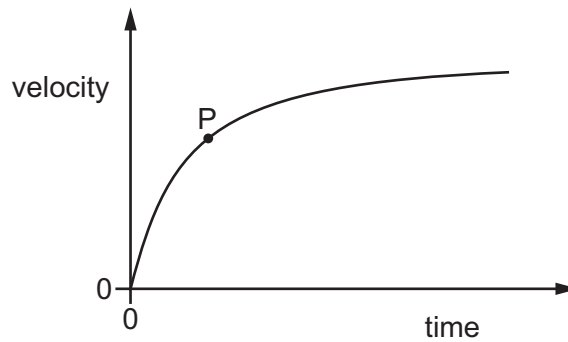
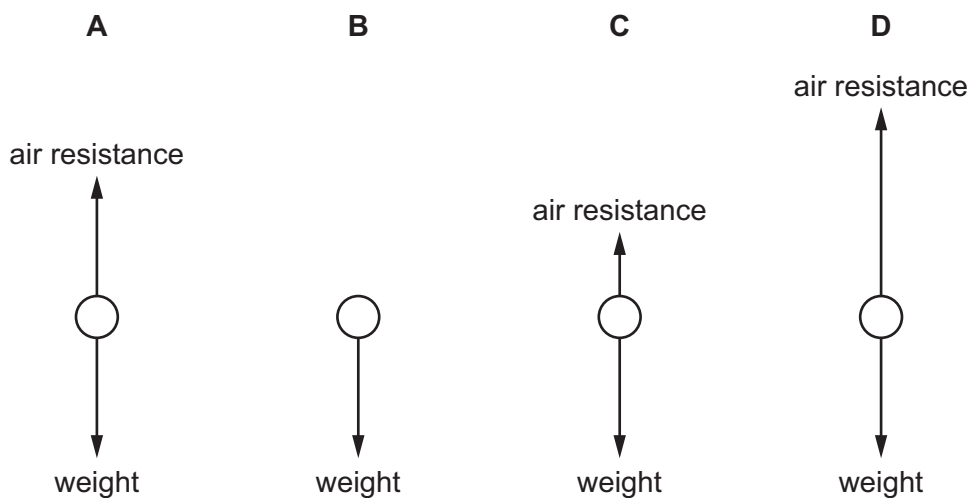


- 8 A sphere falls from rest through the air. The graph shows the variation with time of the sphere's velocity.



Which diagram shows the forces acting on the sphere when it is at the velocity corresponding to point P on the graph?



- 9 In which situation is total linear momentum **always** conserved?
- A in all collisions between two objects
 - B in collisions between two objects moving at equal and opposite velocity
 - C in collisions between two objects that form an isolated system
 - D in collisions between two objects with the same mass
- 10 An object of mass m moving with velocity 9.0 ms^{-1} has a head-on elastic collision with a stationary object of mass $2m$.

After the collision, both objects are moving. No external forces act on the system.

What is the velocity of the object of mass $2m$ after the collision?

- A -6.0 ms^{-1} B -3.0 ms^{-1} C 4.5 ms^{-1} D 6.0 ms^{-1}